

JNK Suppresses Apoptosis via Phosphorylation of the Proapoptotic Bcl-2 Family Protein BAD

Chenfei Yu,¹ Yuzuru Minemoto,¹ Jiyan Zhang,¹
Jing Liu,¹ Fangming Tang, Truc N. Bui,
Jialing Xiang,^{2,*} and Anning Lin^{*}
Ben May Institute for Cancer Research
The University of Chicago
5841 South Maryland Avenue
MC 60627
Chicago, Illinois 60637

Summary

JNK has been suggested to be proapoptotic, antiapoptotic, or have no role in apoptosis depending on the cell type and stimulus used. The precise mechanism of JNK action, under conditions when it promotes cell survival, is not entirely clear. Here, we report that JNK is required for IL-3-mediated cell survival through phosphorylation and inactivation of the proapoptotic Bcl-2 family protein BAD. IL-3 withdrawal-induced apoptosis is promoted by inhibition of JNK but suppressed by expression of a constitutively active JNK. JNK phosphorylates BAD at threonine 201, thereby inhibiting BAD association with the antiapoptotic molecule BCL-X_L. IL-3 induces BAD phosphorylation at threonine 201, and replacement of threonine 201 by alanine generates a BAD mutant, which promotes IL-3 withdrawal-induced apoptosis. Thus, our results provide a molecular mechanism by which JNK contributes to cell survival.

Introduction

The c-Jun N-terminal protein kinase (JNK; also known as stress-activated protein kinase, SAPK) (Hibi et al., 1993) is a member of the mitogen-activated protein kinase (MAPK) superfamily and has been demonstrated to play a critical role in many cellular activities, from growth control to programmed cell death (Chang and Karin, 2001; Davis, 2000; Lin, 2003). JNK is activated by sequential protein phosphorylation through a MAP kinase module, i.e., MAP3K→MAP2K→MAPK (Karin, 1995; Lin, 2003), in response to a variety of extracellular stimuli. Two MAP2Ks (JNKK1/MKK4/SEK1 and JNKK2/MKK7) for JNK have been identified (Derijard et al., 1995; Lin et al., 1995; Lu et al., 1997; Sanchez et al., 1994; Tournier et al., 1997). Although replacement of the conserved Ser/Thr residues with acidic glutamic acids in the T loop of JNKKs failed for unknown reasons to generate constitutively active JNKKs, fusion of JNKK2, which has considerable basal kinase activity, to JNK1 results in a constitutively active c-Jun kinase (Zheng et al., 1999).

Several MAP3Ks, such as MEKK1 (Minden et al., 1994), ASK1 (Ichijo et al., 1997), MLK (Tibbles et al., 1996), TAK1 (Shibuya et al., 1996), and TPL-2 (Salmeron et al., 1996), have been reported to act as MAP3Ks for JNK. While genetic disruption of MEKK1 in mice abrogated JNK activation by certain stimuli (Xia et al., 2000; Yujiri et al., 1998), the physiological relevance of other MAP3Ks in JNK activation is not completely understood. Activation of JNK is also regulated by scaffold proteins, such as JIP, β -arrestin, and JSAP1 (Ito et al., 1999; McDonald et al., 2000; Whitmarsh et al., 1998). The activation of JNK can be terminated by protein phosphatases, such as mitogen-activated protein phosphatases (Karin, 1995), or negatively regulated by NF- κ B-mediated inhibition (De Smaele et al., 2001; Karin and Lin, 2002; Lin, 2003; Tang et al., 2001a, 2002).

Programmed cell death (apoptosis) is essential to many biological processes in multicellular organisms, including embryonic development, immune responses, tissue homeostasis, and normal cell turnover (Jacobson et al., 1997; Kerr et al., 1972; Rinkenberger and Korsmeyer, 1997; Thompson, 1995). The life or death of a cell is determined by the balance between cellular survival factors and death factors. Many growth factors, including the cytokine interleukin 3 (IL-3), are key survival factors that suppress the occurrence of apoptosis. Withdrawal of survival factors, such as IL-3, alone can induce apoptosis (Zha et al., 1996). It has been shown that the survival signal of IL-3 in hematopoietic FL5.12 cells is mediated by phosphorylation of BAD at Ser112 and Ser136 (Zha et al., 1996). The phosphorylation sequesters BAD in the cytoplasm through interaction with 14-3-3, thereby preventing BAD interaction with BCL-2/BCL-X_L on the mitochondrial membrane (Zha et al., 1996). Several protein kinases, such as PKA, Raf-1, or Raf-1-associated kinase(s), Akt/PKB, Rsk2, and CaMKII, have been reported to phosphorylate BAD at one or both of the serines in response to IL-3 or other survival signals (Datta et al., 1997; del Peso et al., 1997; Harada et al., 1999; Kelekar et al., 1997; Wang et al., 1996, 1999; Zha et al., 1996). However, it has been postulated that the survival signal of IL-3 may also be mediated by mechanisms other than phosphorylation of BAD on Ser112 and Ser136 (Hinton and Welham, 1999).

Whereas the initiation and execution of apoptosis depends on activation of the extrinsic- and/or intrinsic-death pathways, the apoptotic process is subjected to regulation by many other signaling pathways, including the JNK pathway. Despite intensive studies, the role of JNK in apoptosis remains highly controversial, as it has been reported to have proapoptotic or antiapoptotic functions or no role in the apoptotic process, depending on the cell type and stimulus used (Lin, 2003). It has been shown that the role of JNK in TNF- α -induced apoptosis is controlled by NF- κ B. TNF- α induces prolonged JNK activation in mouse embryonic fibroblasts deficient in IKK β , which is essential for NF- κ B activation by TNF- α , or RelA, which is the major transcription activity of NF- κ B, or in TNF- α -sensitive human breast carcinoma MCF-7 cells (De Smaele et al., 2001; Tang et al., 2001a,

*Correspondence: xiang@iit.edu (J.X.), alin@huggins.bsd.uchicago.edu (A.L.)

¹These authors contributed equally to this work.

²Present address: Department of Biological, Chemical, and Physical Science, Illinois Institute of Technology, Chicago, Illinois 60616.

2002). The prolonged but not the transient JNK activation contributes to TNF- α -induced apoptosis (Karin and Lin, 2002; Lin, 2003; Tang et al., 2001a, 2002). This conclusion is further supported by the observation that in hepatocyte-specific *Ikk β* deficient mice, concanavalin A (ConA), whose cytotoxic effects are mediated by cell-bound TNF- α , induces prolonged JNK activation, which is required for ConA-induced liver degeneration (Maeda et al., 2003). The contribution of prolonged JNK activation to TNF- α -induced apoptosis is likely by inducing proteolysis of the proapoptotic molecule BID, resulting in a novel form of BID, jBID (Deng et al., 2003). It is proposed that jBID induces the release of Smac (second mitochondria-derived activator of caspase) from mitochondria and that Smac in turn displaces cIAP1 from TRAF2 (TNF-receptor associated factor 2) in the TNF-R1 (TNF-receptor I) complex, thereby relieving the inhibition of cIAP1 on activation of caspase-8 and initiation of apoptosis (Deng et al., 2003). In contrast, the molecular target(s) of JNK action in cell survival is unknown. We report here that JNK activation is required for IL-3-mediated survival through phosphorylation and inactivation of the proapoptotic molecule BAD.

Results

JNK Activation Is Required for IL-3-Mediated Cell Survival

Hematopoietic pro-B FL5.12 cells have successfully been used as a model to study the molecular mechanism underlying growth factor withdrawal-induced apoptosis (Zha et al., 1996). We used FL5.12 cells, either parental or stably expressing BCL-X_L and BAD (Zha et al., 1996), to analyze the role played by JNK in apoptosis. Upon IL-3 withdrawal, both FL5.12 and FL5.12 BCL-X_L/BAD cells underwent apoptosis, which was reversed by IL-3 readdition (Figures 1A and 1B), consistent with the previous report (Zha et al., 1996). Immune complex kinase assays showed that JNK, a protein kinase that is often thought of as a proapoptotic kinase, was inhibited by IL-3 withdrawal but stimulated by IL-3 readdition (Figure 1C). Activation of JNK by IL-3 readdition was significantly blocked by the synthetic JNK inhibitor SP600125 (Figure 1C). The stimulation or inhibition of JNK activity was not the result of changes in its expression, as demonstrated by immunoblotting with anti-JNK antibody (Figure 1C). Similar results were obtained in FL5.12 BCL-X_L/BAD cells (data not shown). These data suggest that JNK is an IL-3-activated protein kinase.

To determine the role of JNK activation by IL-3 in cell survival, FL5.12 cells were subjected to IL-3 withdrawal, treated with or without the JNK inhibitor SP600125, and followed by IL-3 readdition. Apoptotic cell death assays revealed that pretreatment with SP600125 significantly reduced the survival effect conveyed by IL-3 readdition (Figure 1D). Sixty hours after IL-3 withdrawal, cells treated with SP600125 thirty minutes prior to IL-3 readdition had forty percent apoptotic cell death, whereas cells treated with IL-3 readdition in the absence of SP600125 had only five percent apoptotic cell death (Figure 1D). Similar results were obtained in FL5.12 BCL-X_L/BAD cells (data not shown). Furthermore, inhibition of JNK by SP600125 promoted IL-3 withdrawal-induced

apoptosis (Figure 1E). Expression of the dominant-negative JNKK2(K149M) mutant, which suppresses JNK activation (Tang et al., 2001a), also promoted IL-3 withdrawal-induced apoptosis (data not shown). Thus, activation of JNK is involved in IL-3-mediated cell survival.

If JNK is one of the downstream effectors of IL-3, activation of JNK should suppress, at least in part, IL-3 withdrawal-induced apoptosis. To test this scenario, FL5.12 BCL-X_L/BAD cells were stably transfected with mammalian expression vectors encoding HA-JNKK2-JNK1, which has constitutive Jun kinase activity; HA-JNKK2(K149M)-JNK1, which is a kinase-deficient mutant (Zheng et al., 1999; Tang et al., 2001a, 2002); or empty vector. Immune complex kinase assays confirmed that JNKK2-JNK1 had constitutive JNK activity, whereas JNKK2(K149M)-JNK1 had no detectable activity (Figure 1F), although the protein expression level of JNKK2(K149M)-JNK1 was higher than JNKK2-JNK1 (Figure 1F). Apoptotic cell death assays revealed that cells expressing the constitutively active JNKK2-JNK1 were much less sensitive to IL-3 withdrawal-induced apoptosis. Fifty-four hours after IL-3 withdrawal, apoptotic cell death was about forty-three percent and fifty percent in cells stably transfected with empty vector and the kinase-deficient JNKK2(K149M)-JNK1 mutant, respectively (Figure 1G). Under the same conditions, apoptotic cell death was only 15% in cells stably transfected with the constitutively active JNKK2-JNK1 (Figure 1G). There was no detectable difference in cell growth rate among these cell lines (data not shown). In two additional IL-3-dependent MC/9 and FD-CP-1 cells (Scheid et al., 1999), JNK was also activated by IL-3 (Figure 1H), and its inhibition promoted cell death (Figure 1I). Thus, activation of JNK can suppress IL-3 withdrawal-induced apoptosis.

JNK Is an IL-3-Activated BAD Kinase

The survival effect of IL-3 is mainly mediated by phosphorylation and consequent inactivation of the proapoptotic molecule BAD through activation of a group of protein kinases. Withdrawal of IL-3 resulted in hypophosphorylation of BAD, whereas IL-3 readdition induced hyperphosphorylation of BAD at several serine residues, including Ser112 (Figure 2A). Since activation of JNK by IL-3 was involved in survival of IL-3-dependent cells (Figure 1), we hypothesized that JNK may exert its survival effect through phosphorylation of BAD. Indeed, immune complex kinase assays showed that IL-3-activated JNK significantly phosphorylated GST-BAD in a manner that mirrored the phosphorylation pattern of BAD in vivo (compare Figure 2B to 2A). Phosphorylation of GST-BAD by JNK occurred on BAD, as GST was not a JNK substrate (Lin et al., 1995). The ability of JNK to phosphorylate GST-BAD was well correlated to its phosphorylation of GST-c-Jun, an authentic JNK substrate (Figure 2B). Additionally, UV-activated JNK also phosphorylated GST-BAD (Figure 2C).

To test whether JNK directly phosphorylates BAD, purified wild-type (wt) GST-JNK1 was incubated with extracts from cells transfected with a constitutively active MEKK1 (Lin et al., 1995), in a kinase reaction with nonradioactive ATP, isolated by glutathione-Sepharose

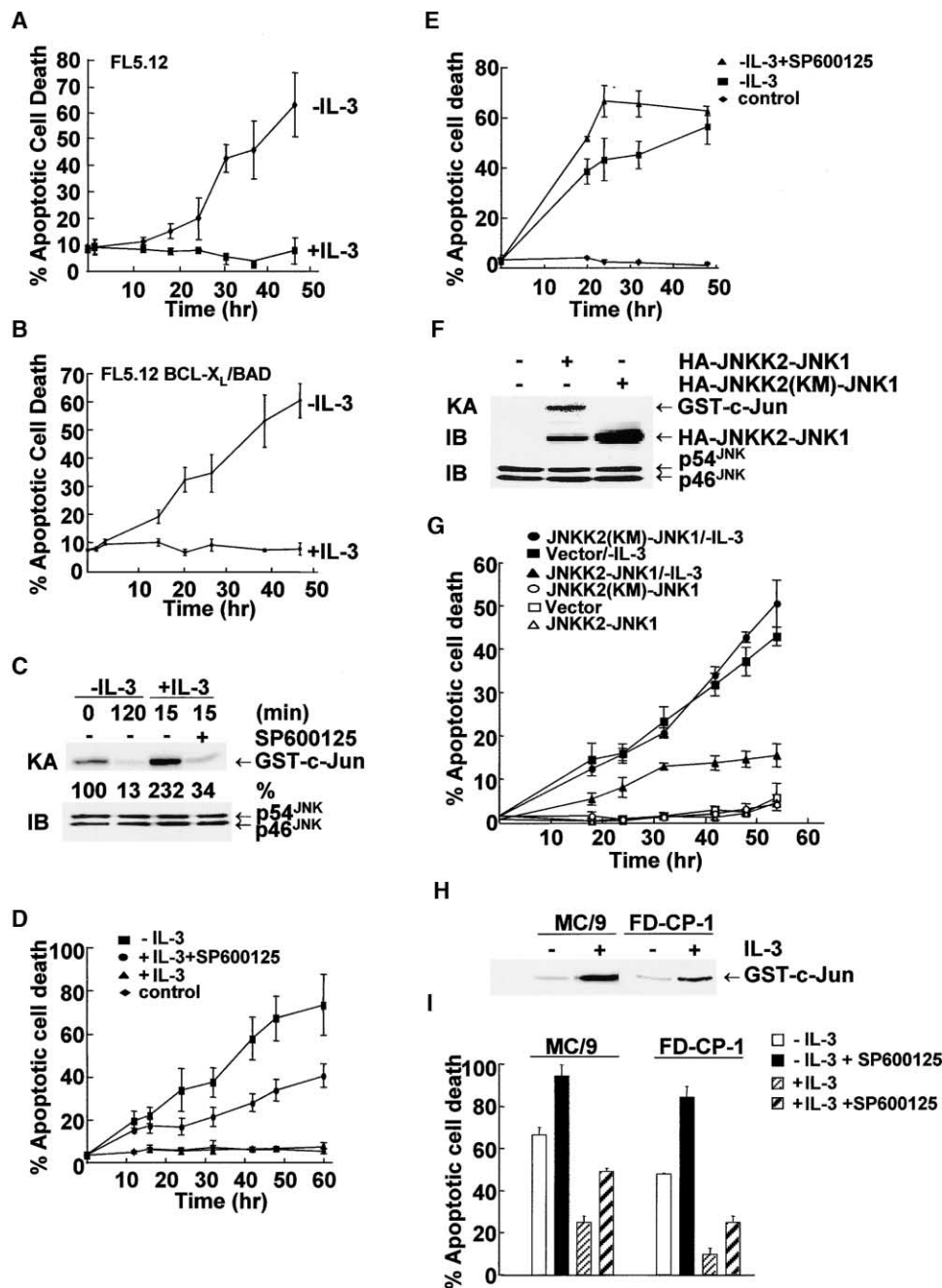


Figure 1. JNK Activation Is Required for IL-3-Mediated Cell Survival

FL5.12 cells (A) and FL5.12 BCL-X_L/BAD (B) cells were deprived of IL-3 for 2 hr (-IL-3), followed by IL-3 readdition (+IL-3). Apoptotic cell death was measured with Hoechst 33258 staining. The results are presented as means $\pm$ standard error and represent three individual experiments. (C) FL5.12 cells were deprived of IL-3 for 2 hr, followed by IL-3 readdition for 15 min. The JNK inhibitor SP600125 (10 μ M) was added 30 min prior to IL-3 readdition. The activity and expression level of JNK was measured as described (Lin et al., 1995). (D and E) FL5.12 cells were deprived of IL-3 for 2 hr (-IL-3), followed by IL-3 readdition in the presence or absence of SP600125 (10 μ M) (+IL-3 and +IL-3 + SP600125, respectively) (D), or left untreated in the presence or absence of SP600125 (10 μ M) (-IL-3 and -IL-3/SP600125) (E). SP600125 (10 μ M) was added 30 min prior to IL-3 withdrawal. Apoptotic cell death was analyzed as described in (A). (F) HA-JNKK2-JNK1 and JNKK2(KM)-JNK1 were immunoprecipitated with anti-HA antibody from FL5.12 BCL-X_L/BAD cells that stably express HA-JNKK2-JNK1 or HA-JNKK2(KM)-JNK1, respectively. The activity and expression level of HA-JNKK2-JNK1 or HA-JNKK2(KM)-JNK1 were measured as described (Zheng et al., 1999). (G) Stable FL5.12 BCL-X_L/BAD clones were deprived of IL-3 for 2 hr, followed by readdition of IL-3 (+IL-3), or left untreated (-IL-3), as indicated. Apoptotic cell death was analyzed as described in (A). (H) MC/9 and FD-CP-1 cells were stimulated with or without IL-3 for 15 min and JNK activity was measured as described in (C). (I) MC/9 and FD-CP-1 cells were deprived of IL-3 for 4 hr, followed by addition of IL-3, or left alone in the presence or absence of SP600125 (10 μ M) for 24 hr (MC/9) or 10 hr (FD-CP-1). Apoptotic cell death was analyzed as in (A).

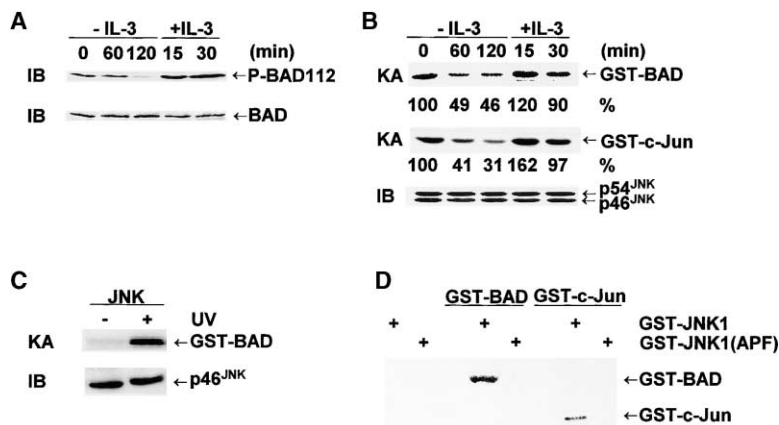


Figure 2. JNK Is an IL-3-Activated BAD Kinase

(A and B) FL5.12 BCL-X_L/BAD cells were deprived of IL-3 for 1 or 2 hr, followed by IL-3 readdition for 15 min or 30 min. (A) S112 phosphorylation and expression of BAD were detected by immunoblotting using anti-phospho-S112 antibody (upper panel) or anti-BAD antibody (lower panel). (B) Phosphorylation of GST-BAD (upper panel) or GST-c-Jun (1-79) (middle panel) by IL-3-activated JNK was measured by immune complex kinase assays. (C) HeLa cells were treated with or without UV (60 J/m²). Phosphorylation of GST-BAD by UV-activated JNK was measured as in (B). (D) Phosphorylation of GST-BAD or GST-c-Jun by activated recombinant GST-JNK1 in vitro (Lin et al., 1995). Inactive GST-JNK1(APF) was used as a control.

(GSH) beads, and extensively washed. The isolated GST-JNK1 was active, as it significantly phosphorylated GST-c-Jun (Figure 2D). The active GST-JNK1 also phosphorylated GST-BAD (Figure 2D). However, it is possible that other kinases activated by MEKK1 and/or JNK might be present in the reaction and mediate BAD phosphorylation. In this scenario, these kinases should have already been activated prior to binding to GST-JNK1 proteins, as cells expressing MEKK1 have activated endogenous JNKK and JNK. This is unlikely. Under the same conditions the kinase-deficient GST-JNK1(APF), in which threonine 183 and tyrosine 185 have been replaced by nonphosphorylatable alanines and therefore can no longer be activated (Figure 2D), was unable to phosphorylate GST-BAD (Figure 2D). Thus, these results show that JNK is an IL-3-activated BAD kinase.

JNK Phosphorylates BAD at Threonine 201 In Vitro

To identify JNK phosphorylation site(s) in BAD, basal JNK was isolated from IL-3-deprived FL5.12 BCL-X_L/M2-BAD or resting HeLa cells, whereas active JNK was isolated from IL-3-stimulated FL5.12 BCL-X_L/M2-BAD or UV-irradiated HeLa cells. Two-dimensional tryptic phosphopeptide mapping revealed that basal JNK-phosphorylated GST-BAD contained a minor phosphopeptide *a* (Figures 3A and 3B), whereas active JNK-phosphorylated GST-BAD contained a major phosphopeptide *b* in addition to the phosphopeptide *a* (Figures 3A and 3B). This suggests that phosphopeptide *b* was specifically phosphorylated by active JNK. Phosphoamino acid analysis revealed that GST-BAD phosphorylated by basal JNK contained only phosphoserine (PS). In contrast, GST-BAD phosphorylated by active JNK mainly contained phosphothreonine (PT) with only a marginal amount of phosphoserine (PS) (Figure 3C). Furthermore, the phosphopeptide *b* contained only phosphothreonine (Figure 3C). These data demonstrate that BAD is phosphorylated by active JNK at threonine residue(s).

To locate JNK-phosphorylated threonine residue(s), we constructed a C-terminal truncated GST-ΔC-BAD (1-114) and an N-terminal truncated GST-ΔN-BAD(115-204) (Figure 3D). Immune complex kinase assays showed that only GST-ΔN-BAD was phosphorylated by JNK (Figure 3E), although a similar amount of both truncated

GST-BAD proteins was used in the kinase reaction (Figure 3E). Two-dimensional tryptic phosphopeptide mapping revealed that GST-ΔN-BAD contained two phosphopeptides, *a'* and *b'* (Figure 3F, left two panels), which had identical electrophoretic and chromatographic mobility to those of phosphopeptides *a* and *b* from the phosphorylated full-length GST-BAD (Figure 3F, right two panels). This suggests that the JNK phosphorylation site(s) is located in the C-terminal half of BAD.

As a MAP kinase, JNK typically phosphorylates its substrates at either Ser or Thr residue, which is immediately followed by a Pro residue (Hibi et al., 1993; Karin, 1995). Although the ΔN-BAD mutant contains multiple Thr residues, only Thr201 was followed by a Pro residue. Therefore, Thr201 was replaced by nonphosphorylatable alanine (Ala) in the full-length GST-BAD using site-directed mutagenesis (Figure 4A). Immune complex kinase assays showed that active JNK only phosphorylated wt GST-BAD but not the GST-BAD(T201A) mutant (Figure 4B). Since GST-BAD(T210A) can still be phosphorylated by active Akt to a similar extent when compared to wt GST-BAD (Figure 4C), the inability of active JNK to phosphorylate GST-BAD(T201A) was not caused by potential changes in the global structure of GST-BAD(T201A) protein. Two-dimensional tryptic phosphopeptide mapping revealed that replacement of Thr201 by Ala completely eliminated the phosphopeptide *b* but had no effect on phosphopeptide *a* (Figure 4D). Consistently, phosphoamino acid analysis showed that the GST-BAD(T201A) mutant contained only phosphoserine (PS) (Figure 4E), whereas wt GST-BAD contained both Ser and Thr (Figure 3C). The observation that JNK phosphorylates BAD at Thr201 was further confirmed by Edman degradation. The hydrolysis of the phosphopeptide *b* resulted in the accumulation of free phosphate in the fourth cycle, consistent with the prediction that Thr201 is the phosphorylated amino acid in the phosphopeptide *b* (GGSpTPSE) (data not shown). Thus, BAD is phosphorylated by active JNK at Thr201.

Activation of the JNK Signaling Pathway Leads to Phosphorylation of BAD at Thr201 In Vivo

To determine whether JNK activation is able to induce phosphorylation of BAD at Thr201 in vivo, we generated a rabbit polyclonal antibody using a synthetic phospho-

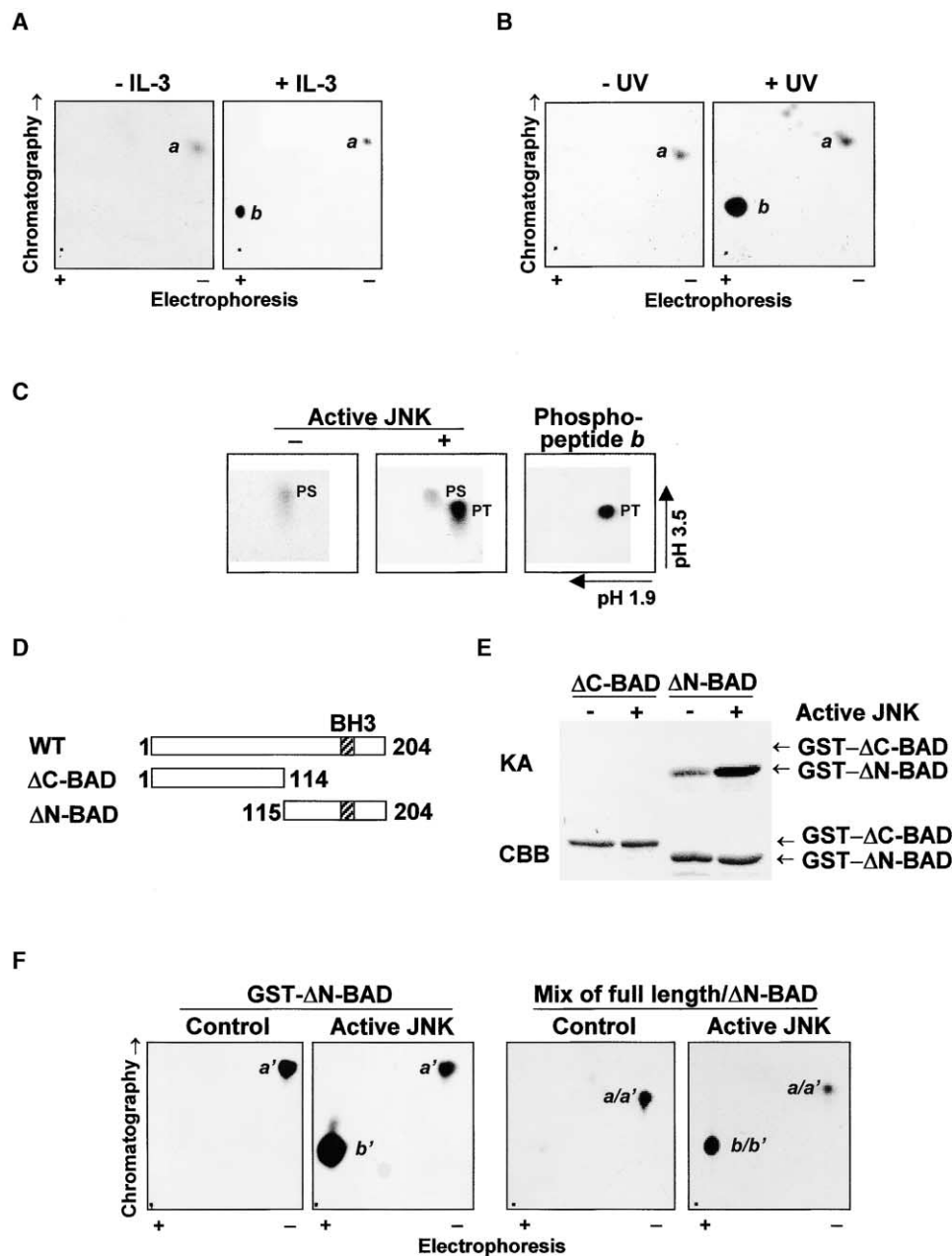


Figure 3. BAD Is Phosphorylated at Threonine Residue by JNK

(A and B) GST-BAD proteins phosphorylated by either IL-3 or UV-activated JNK were subjected to two-dimensional tryptic phosphopeptide mapping, as described (Lin et al., 1992). Spot *a* is the tryptic peptide phosphorylated by basal JNK; spot *b* represents the peptide phosphorylated by activated JNK.

(C) Phosphorylated BAD or the phosphopeptide *b* were subjected to phosphoamino acid analysis, as described (Lin et al., 1992). PS, phosphoserine; PT, phosphothreonine.

(D) Schematic presentation of truncated BAD constructs. ΔC-BAD, a truncated BAD, contains amino acids 1 to 114, whereas ΔN-BAD contains 115 to 204.

(E) GST-ΔN-BAD but not GST-ΔC-BAD was phosphorylated by UV-activated JNK as in Figure 2C. CBB, Coomassie brilliant blue staining.

(F) Two-dimensional tryptic phosphopeptide mapping of phosphorylated GST-ΔN-BAD proteins (left panel) and mixed phosphorylated full-length and ΔN-BAD proteins (right panel).

peptide containing phosphorylated BAD Thr201 as an immunogen. Immunoblotting analysis showed that anti-phospho-Thr201 antibody specifically recognized active JNK-phosphorylated GST-BAD but not nonphosphorylated GST-BAD (Figure 5A, upper panel). Under the same

conditions, anti-phospho-Thr201 antibody was unable to recognize the GST-BAD(T201A) mutant (Figure 5A, upper panel). This was not a result of the difference in the amount of GST-BAD proteins, as analyzed by immunoblotting with anti-GST antibody (Figure 5A,

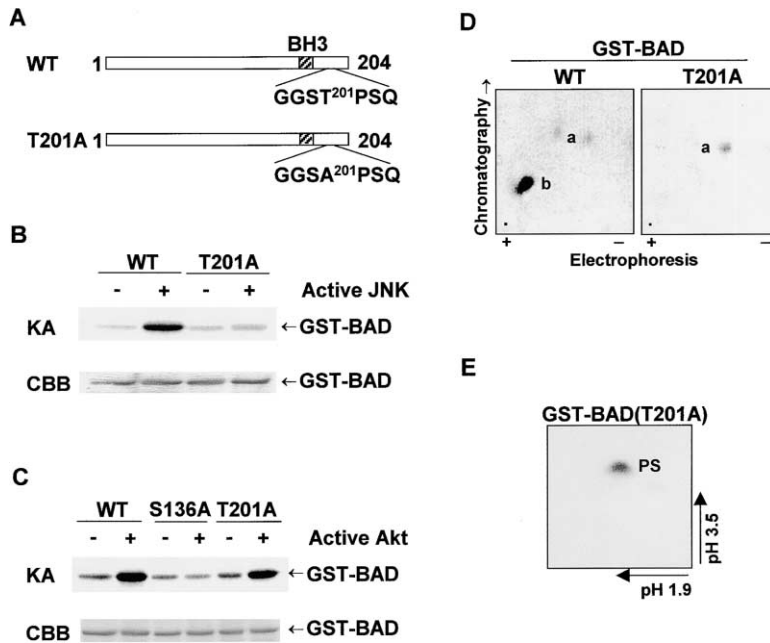


Figure 4. BAD Is Phosphorylated by JNK at Thr201

(A) Schematic presentation of the BAD (T201A) mutant. GGST²⁰¹PSQ corresponds to the phosphopeptide *b*. (B) Replacement of Thr201 in GST-BAD abolished UV-activated JNK-mediated phosphorylation of BAD. (C) The GST-BAD(T201A) mutant was still phosphorylated by active Akt. (D) Two-dimensional tryptic phosphopeptide mapping of active JNK-phosphorylated GST-BAD and GST-BAD(T201A) proteins. (E) Phosphoamino acid analysis of phosphorylated GST-BAD(T201A). PS, phosphoserine.

lower panel). Thus, anti-phospho-Thr201 antibody specifically recognizes BAD when it is phosphorylated at Thr201 residue.

To determine whether JNK activation is required for BAD phosphorylation at Thr201, we used JNK1-deficient mice embryonic fibroblasts, in which JNK activation by

10% fetal bovine serum (FBS) (Figure 5B, KA panel) or EGF (data not shown) was undetectable. Immunoblotting using anti-phospho-Thr201 antibody revealed that FBS-treatment induced BAD phosphorylation at Thr201 in wt but not in JNK1 null fibroblasts, although expression levels of BAD were similar in both cells (Figure 5B).

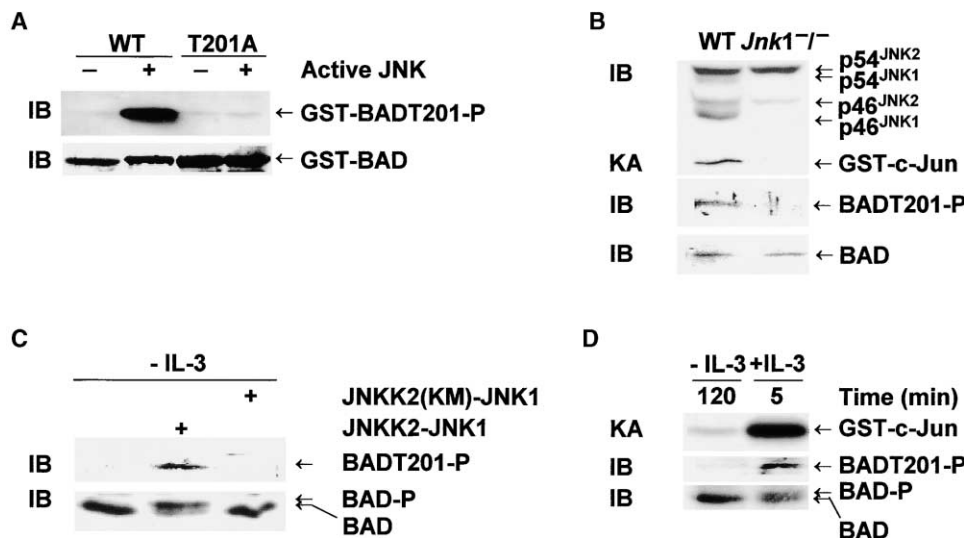


Figure 5. JNK Is Necessary and Sufficient to Phosphorylate BAD at Thr201 In Vivo

(A) GST-BAD and GST-BAD(T201A) were phosphorylated by active JNK in the presence of nonradioactive ATP (17 μ M) and analyzed by immunoblotting using anti-phospho-Thr201 antibody (upper panel). The amount of GST-BAD and GST-BAD(T201A) was examined by immunoblotting using anti-GST antibody (lower panel). (B) wt and JNK1 null MEFs were treated with 10% FBS. The expression and activity of JNK was measured (upper and upper middle panels, respectively), as described in Figure 1C. Thr201 phosphorylation and expression of BAD were analyzed by immunoblotting with anti-phospho-Thr201 and anti-BAD antibodies (lower middle and lower panels, respectively). (C) FL5.12 BCL-X_i cells were transfected with expression vector encoding M2-BAD along with HA-JNKK2-JNK1, HA-JNKK2(K149M)-JNK1, or empty vector. After 24 hr, cells were deprived of IL-3 for 2 hr. Thr201 phosphorylation and expression of BAD were analyzed as described in (B). BAD-P, upshifted phosphorylated BAD. (D) FL5.12 BCL-X_i/M2-BAD cells were deprived of IL-3 for 2 hr (-IL-3), followed by IL-3 readdition (+IL-3) for 5 min. Activation of JNK was determined by immune complex kinase assays (upper panel). Thr201 phosphorylation and expression of BAD were analyzed as described in (B).

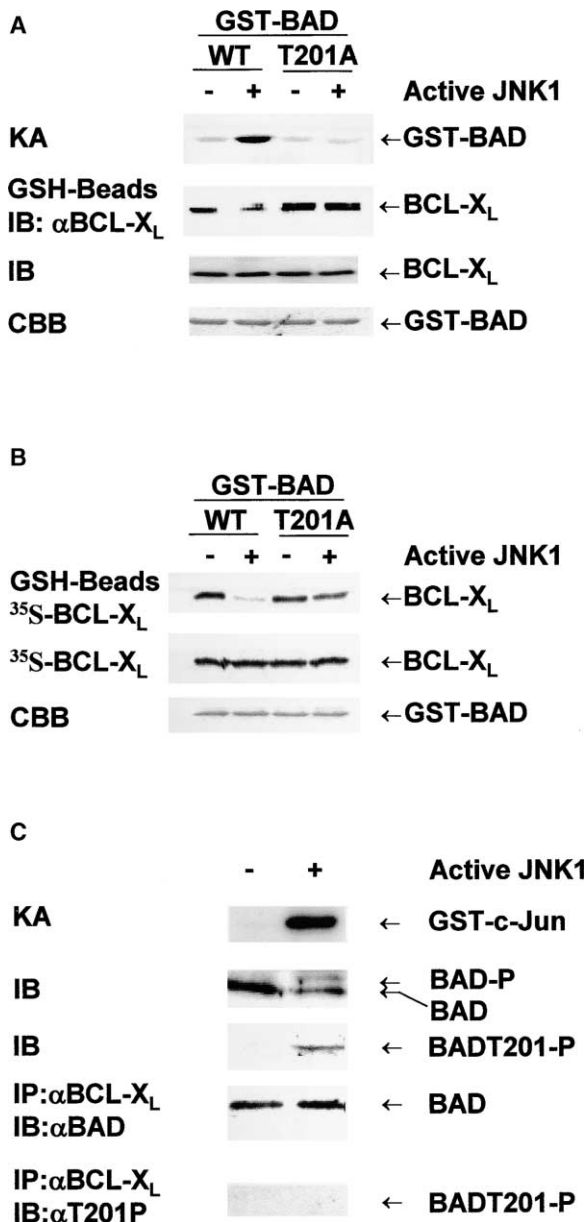


Figure 6. Phosphorylation of BAD by JNK at Thr201 Inhibits Its Association with BCL-X_L In Vitro and In Vivo

Purified wt GST-BAD and GST-BAD(T201A) were incubated with active JNK in the presence of either [γ -³²P] ATP (upper panel) or 17 μ M nonradioactive ATP (upper middle panel). Modified GST-BAD proteins were isolated with GSH beads and mixed with either FL5.12 Bcl-X_L cell extracts (1 mg) (A) or in vitro translated ³⁵S-labeled BCL-X_L (B) for 12 hr and extensively washed. GST-BAD-associated BCL-X_L (upper middle panel) or BCL-X_L in one-tenth of input cell extracts (lower middle panel) was analyzed by immunoblotting using anti-BCL-X_L antibody. The amount of wt GST-BAD and GST-BAD(T201A) was examined by CBB staining (lower panel). GST-BAD-associated ³⁵S-BCL-X_L was visualized by radioautograph (B). (C) FL5.12 BCL-X_L cells were transfected with expression vector encoding M2-BAD along with HA-JNKK2-JNK1 or empty vector. After 24 hr, cells were deprived of IL-3 for 2 hr. The activity of HA-JNKK2-JNK1 (upper panel) was determined as in Figure 1F. Expression and Thr201 phosphorylation of BAD were analyzed (upper middle and middle panels, respectively), as in Figure 5C. BAD-P, upshifted phosphorylated BAD. BCL-X_L was immunoprecipitated with anti-BCL-X_L antibody,

Thus, JNK activation is necessary for BAD phosphorylation at Thr201.

We next tested whether activation of JNK is sufficient to induce BAD phosphorylation at Thr201. In FL5.12 BCL-X_L cells, expression of the constitutively active JNKK2-JNK1 but not the kinase-deficient JNKK2(K149M)-JNK1 resulted in phosphorylation of cotransfected M2-BAD at Thr201 in the absence of IL-3, as revealed by immunoblotting with anti-phospho-Thr201 antibody (Figure 5C). Interestingly, phosphorylation at Thr201 caused slow migration, an upshift, of BAD on SDS gel during electrophoresis (Figure 5C), similar to Ser112/Ser136 phosphorylated BAD (Zha et al., 1996; Wang et al., 1996). Thus, activation of JNK is sufficient for BAD phosphorylation at Thr201 in vivo.

Since JNK functions as a downstream effector in the IL-3 signaling pathway, IL-3 treatment should also induce BAD phosphorylation at Thr201. Indeed, immunoblotting with anti-phospho-Thr201 antibody showed that BAD was specifically phosphorylated at Thr201 in response to IL-3 (Figure 5D, middle panel). The phosphorylation of BAD at Thr201 occurred as early as 5 min after IL-3 readdition, corresponding to the initiation of JNK activation by IL-3 (Figure 5D, upper panel). Taken together, BAD phosphorylation by JNK at Thr201 may be involved in the IL-3 signaling for cell survival.

Phosphorylation of BAD by JNK at Thr201 Decreases Its Association with BCL-X_L

Phosphorylation of BAD at several Ser residues (112, 136, and 155) resulted in inhibition of the proapoptotic activity of BAD and apoptosis, by reducing BAD association with the antiapoptotic molecule BCL-X_L. We hypothesized that phosphorylation of BAD by JNK at Thr201 may also reduce BAD association with BCL-X_L. To test this idea, wt GST-BAD or GST-BAD(T201A) proteins were subjected to phosphorylation by active JNK1. As expected, wt GST-BAD but not the GST-BAD(T201A) was phosphorylated by active JNK1 (Figure 6A, upper panel). In parallel, GST-BAD proteins were phosphorylated by active JNK1 in the presence of nonradioactive ATP, isolated by GSH bead, and mixed with cell extracts containing BCL-X_L. GST pull-down in combination with immunoblotting using anti-BCL-X_L antibody revealed that phosphorylation by JNK significantly reduced the association of wt GST-BAD and BCL-X_L (Figure 6A, upper middle panel). In contrast, JNK has no detectable effect on the association of GST-BAD(T201A) mutant with BCL-X_L (Figure 6A, upper middle panel). This was not the result of variations of BCL-X_L or GST-BAD in the reaction, as demonstrated by immunoblotting with anti-BCL-X_L antibody and Commassie brilliant blue (CBB) staining of GST-BAD proteins (Figure 6A, lower middle and lower panel). To exclude the possibility that phosphorylation of BAD by JNK may affect BAD association with unknown proteins involved in the BAD/BCL-X_L association, we examined GST-BAD association with in vitro-translated, ³⁵S-labeled BCL-X_L. GST pull-down

and BCL-X_L-associated BAD proteins were detected by immunoblotting using anti-BAD (lower middle panel) or anti-phospho-Thr201 antibody (lower panel).

assays revealed that phosphorylation by active JNK1 significantly reduced the binding of wt GST-BAD but not GST-BAD(T201A) to ³⁵S-labeled BCL-X_L (Figure 6B, upper panel). Taken together, phosphorylation of BAD by JNK at Thr201 reduces the binding of BAD to BCL-X_L in vitro.

We next examined whether phosphorylation of BAD at Thr201 affects its association with BCL-X_L in vivo. FL5.12 BCL-X_L cells were transfected with M2-BAD along with HA-JNKK2-JNK1 or empty vector. After 24 hr, cells were deprived of IL-3 for 2 hr. The JNKK2-JNK1 was active under these conditions, as measured by immune complex kinase assays (Figure 6C, upper panel). Immunoblotting with anti-M2 antibody showed that M2-BAD migrated as a single band in cells transfected with empty vector but as a doublet in cells cotransfected with the JNKK2-JNK1 (Figure 6C, upper middle panel). Immunoblotting with anti-phospho-Thr201 revealed that only the upper band of M2-BAD was phosphorylated at Thr201 (Figure 6C, middle panel). When cell extracts were immunoprecipitated using anti-BCL-X_L antibody, only the lower band of M2-BAD, which was not phosphorylated at Thr201 (Figure 6C, middle panel), was coprecipitated with BCL-X_L, as analyzed by immunoblotting with anti-M2 antibody (Figure 6C, lower middle panel). Immunoblotting with anti-phospho-Thr201 antibody revealed that there was no detectable Thr201 phosphorylated M2-BAD coprecipitated with BCL-X_L (Figure 6C, lower panel). These data suggest that phosphorylation of BAD at Thr201 reduced its association with BCL-X_L in vivo.

Elimination of Thr201 Phosphorylation Promotes the Proapoptotic Activity of BAD

To determine whether BAD phosphorylation at Thr201 regulates the proapoptotic activity of BAD in vivo, FL5.12 BCL-X_L clones stably expressing wt-BAD or the BAD(T201A) mutant were used to determine their susceptibility to IL-3 withdrawal-induced apoptosis. Immunoblotting showed that expression of the BAD(T201A) mutant was similar to that of wt-BAD (Figure 7A). However, cells expressing the BAD(T201A) mutant were more sensitive to IL-3 withdrawal-induced apoptosis than cells expressing wt-BAD, as analyzed by apoptotic death assays (Figure 7B). Thirty-six hours after IL-3 withdrawal, cells expressing the BAD(T201A) mutant had fifty percent to sixty percent apoptotic cell death, about twenty to thirty percent more than cells expressing wt-BAD (Figure 7B). Under the same conditions, cells expressing either BAD(S112A) or BAD(S136A) had 50% apoptotic cell death (data not shown; Zha et al., 1996). Thus, the BAD(T201A) mutant has similar potency in promoting apoptosis to the BAD(S112A) or BAD(S136A) mutant (Zha et al., 1996). Consistently, IL-3 withdrawal-induced higher activation of caspase-9 and caspase-3 in cells expressing the BAD(T201A) mutant than cells expressing wt-BAD (Figures 7C and 7D). Furthermore, although overexpression of the JNKK2-JNK1 fusion protein suppressed IL-3 withdrawal-induced apoptosis in FL5.12 BCL-X_L/BAD cells (Figure 1B), it failed to do so in FL5.12 BCL-X_L/BAD(T201A) cells (data not shown). Taken together, these data indicate that phosphorylation of BAD by JNK at Thr201 is involved in IL-3-mediated survival of FL5.12 cells.

To test whether Thr201 phosphorylation works in concert with other serine phosphorylations in suppressing the proapoptotic activity of BAD, we generated a BAD(S112A/T201A) double mutant, in which both Ser112 and Thr201 were replaced with nonphosphorylatable alanines. FL5.12 BCL-X_L cells were transiently transfected with expression vector encoding various M2-BAD constructs (wt, S112A, T201A, and S112A/T201A) along with green fluorescent protein (GFP) at a ratio of 4:1. Immunoblotting analysis with anti-M2 antibody showed that expression levels of M2-BAD proteins were similar (Figure 7E). Apoptotic cell death assays revealed that the BAD(T201A) and the BAD(S112A) mutants had similar proapoptotic potency (Figure 7F). Twenty-four hours after IL-3-withdrawal, cells expressing either BAD(T201A) or BAD(S112A) had about forty percent apoptotic cell death, whereas cells expressing wt-BAD have only twenty-four percent apoptotic cell death (Figure 7F). Interestingly, cells expressing the BAD(S122A/T201A) double mutant were more sensitive to IL-3 withdrawal-induced apoptosis than cells expressing either the BAD(S112A) or BAD(T201A) mutant, with 53% apoptotic cell death, even BAD(S112A/T201A) expression level was slightly less (Figure 7E). These data suggests that Thr201 phosphorylation may work cooperatively with other serine phosphorylations of BAD, such as Ser112 phosphorylation, to inhibit the proapoptotic activity of BAD.

Discussion

It has been proposed that JNK may be involved in cell survival under certain circumstances (Davis, 2000; Minden et al., 1994). Genetic disruption of *jnk1* and *jnk2* alleles in mice revealed that apoptosis was increased in the hindbrain and forebrain regions at E10.5 (Kuan et al., 1999; Sabapathy et al., 1999), suggesting that JNK may be required for survival of cells in certain brain regions during neuronal development. Inhibition of JNK2 activity also resulted in suppression of tumorigenesis through promoting apoptosis (Bost et al., 1999; Chen et al., 2001; Hess et al., 2002; Potapova et al., 2002). However, loss-of-function studies are unable to reveal whether JNK is directly or indirectly involved in cell survival and how JNK promotes cell survival. In this report, we demonstrate that one of the molecular mechanisms by which JNK suppresses apoptosis is through phosphorylation of BAD at Thr201, thereby inhibiting its proapoptotic activity. This conclusion is based on the following evidence.

First, JNK was activated by IL-3, which is a survival cytokine for hematopoietic cells (Figure 1C). Second, the JNK inhibitor SP600125 suppressed IL-3-mediated cell survival and promoted IL-3 withdrawal-induced cell death (Figures 1D and 1E). Similarly, inhibition of JNK by the dominant-negative JNKK2(K149M) mutant promoted IL-3 withdrawal-induced apoptosis (data not shown). Third, expression of the constitutively active JNKK2-JNK1 but not the kinase-deficient JNKK2(K149M)-JNK1 (Figure 1F) (Tang et al., 2001a; Zheng et al., 1999) inhibited IL-3 withdrawal-induced apoptosis (Figure 1G). Fourth, JNK phosphorylated and inactivated the proapoptotic molecule BAD (Figures 2B, 6, and 7). Taken

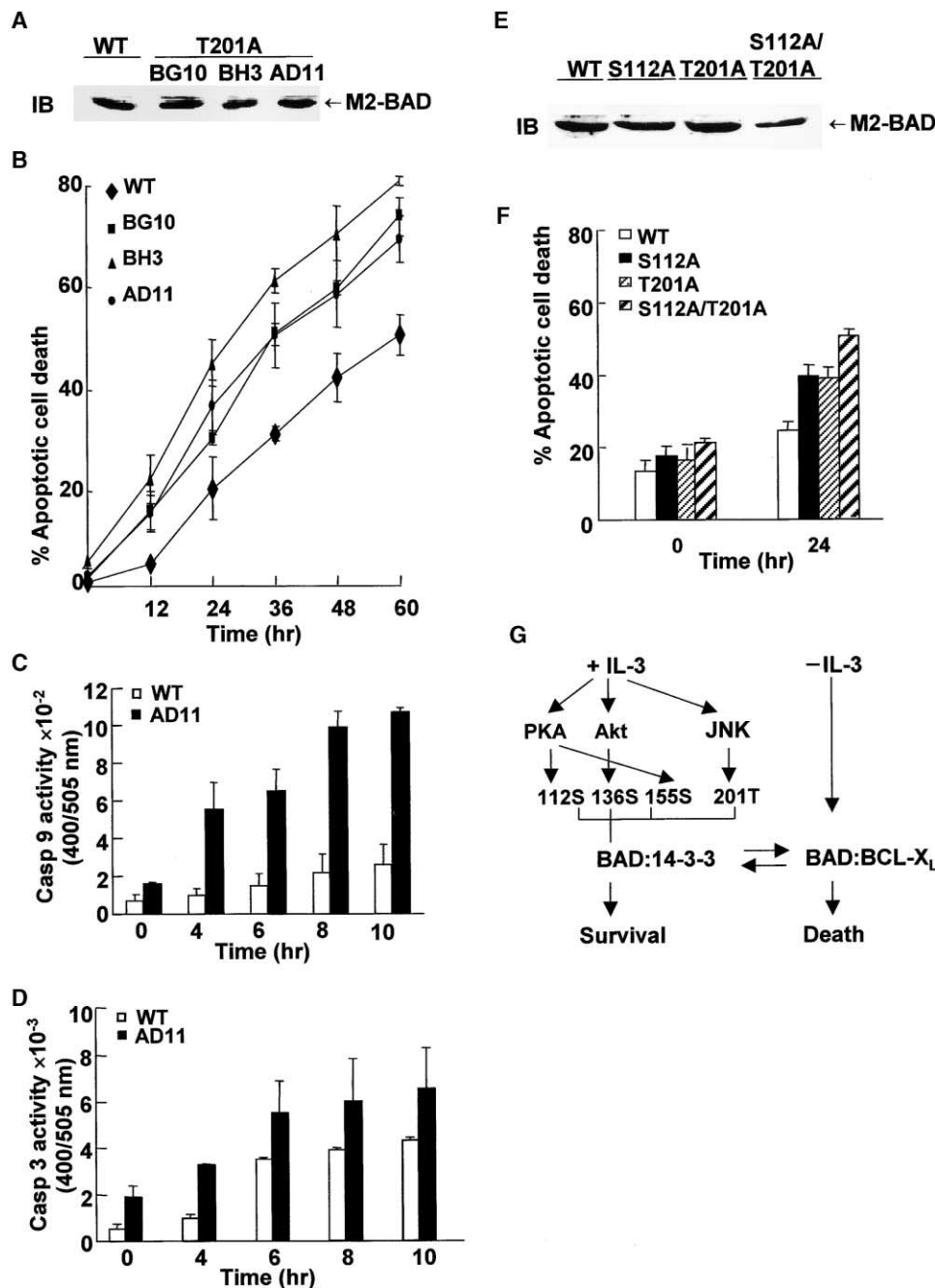


Figure 7. Elimination of Thr201 Phosphorylation Augments BAD Proapoptotic Activity

(A) Immunoblotting analysis of BAD (wt or T201A) in cell extracts from corresponding stable cell lines with anti-M2 antibody.

(B) FL5.12 BCL-X_L/BAD(T201A) cells were more sensitive to IL-3 withdrawal-induced apoptosis. Apoptotic cell death was analyzed as described in Figure 1.

(C and D) Measurement of caspase-9 and caspase-3 activities in extracts of IL-3-deprived FL5.12 BCL-X_L/BAD(T201A) stable cells, respectively (Tang et al., 2001a, 2002).

(E) FL5.12 cells were transfected with expression vectors encoding various M2-BAD constructs (wt, S112A, T201A, and S112A/T201A, 10 μ g each), along with BCL-X_L (10 μ g) and GFP (5 μ g). Expression of various BAD proteins was analyzed by immunoblotting using anti-M2 antibody.

(F) IL-3 withdrawal-induced apoptotic cell death of various transfected, GFP-positive cells in (E).

(G) A schematic presentation of the role of JNK in IL-3-mediated cell survival.

together, these results demonstrate that JNK functions as an antiapoptotic molecule to suppress IL-3 withdrawal-induced apoptosis in hematopoietic cells.

IL-3 exerts its survival activity through activating sev-

eral protein kinases, such as PKA and Akt, which phosphorylate and inactivate the proapoptotic molecule BAD. Our results identified JNK as an IL-3-activated BAD kinase (Figures 1C, 2B, and 2D). Unlike other BAD

kinases that phosphorylate BAD at serine residues, JNK phosphorylated BAD at Thr201 (Figures 3, 4, and 5). Surprisingly, when BAD was metabolically labeled with [32 P] orthophosphate in IL-3-stimulated cells, it was predominantly phosphorylated at serine residues, whereas its threonine phosphorylation was undetectable by phosphoamino acid analysis (data not shown; Zha et al., 1996). One of the possibilities is that Thr201 phosphorylation may be prone to rapid dephosphorylation in vivo. Although we cannot formally exclude this possibility, this may not be likely, as a cocktail of phosphatase inhibitors was used in the in vivo labeling experiments. Another possibility is that Thr201 phosphorylation may only account for a very small portion of the overall phosphorylation of BAD induced by IL-3, which makes its detection technically difficult among overwhelmingly serine phosphorylation. In support of this idea, the phospho-Thr201 antibody was able to detect Thr201 phosphorylation of BAD in cells either expressing the constitutively active JNKK2-JNK1 or treated with IL-3 (Figures 5C, 5D, and 6C).

Although JNK phosphorylates BAD in vitro and in vivo, coimmunoprecipitation experiments were unable to detect their association in vivo (data not shown). This is not a surprise, since the association between JNK and cellular c-Jun was also undetectable by coimmunoprecipitation (data not shown; Hibi et al., 1993). It is likely that the association between JNK and its substrates is highly transient and/or not strong enough to be detected by coimmunoprecipitation.

Our findings that JNK phosphorylated BAD at Thr201 is contradictory to a recent report, which showed that JNK phosphorylated BAD at Ser128, resulting in promotion of the apoptotic effect of BAD in primary neurons (Donovan et al., 2002). However, the only evidence that BAD was phosphorylated by JNK at Ser128 was based on mass spectrometric analysis (Donovan et al., 2002). In contrast, phosphoamino acid analysis revealed that BAD was phosphorylated at threonine, rather than serine, by active JNK (Figures 3 and 4). Consistently, the phosphopeptide *b*, which represents JNK phosphorylated peptide of BAD (Figures 3A and 3B), contained only phosphothreonine (Figure 3C). Furthermore, manual Edman degradation of the phosphopeptide *b* (GGSTPSE) resulted in the release of 32 P in the fourth cycle (data not shown). Most importantly, mutation of Thr201 abolished phosphorylation of the peptide *b* by active JNK (Figure 4D). In contrast, mutation of Ser128 had no effect on JNK-mediated phosphorylation of BAD and two-dimensional phosphopeptide mapping of the BAD(S128A) mutant resulted in identical phosphorylated tryptic peptides *a* and *b* when compared to wild-type BAD (data not shown). Thus, BAD is phosphorylated by JNK at Thr201, rather than Ser128.

Is phosphorylation of BAD by JNK at Thr201 critical for IL-3-mediated cell survival? Previously, Korsmeyer and colleagues have elegantly shown that IL-3-induced phosphorylation of BAD at Ser112 and Ser136 results in inhibition of BAD proapoptotic activity (Zha et al., 1996). Replacement of Thr201 by nonphosphorylatable alanine significantly enhanced IL-3 withdrawal-induced apoptosis (Figure 7), suggesting that Thr201 phosphorylation is also involved in IL-3-mediated cell survival. The BAD(T201A) mutant was as potent as the BAD(S112A)

(Figure 7F) or BAD(S136A) mutant (data not shown) in promoting apoptosis. In addition, the BAD(S112A/T201A) double mutant was more potent than the BAD single mutant in promoting IL-3 withdrawal-induced apoptosis, indicating that Thr201 phosphorylation may work cooperatively with Ser112 (or maybe other serine) phosphorylation in IL-3-mediated cell survival. Therefore, it is likely that the survival signal of IL-3 is mediated by phosphorylation of BAD at both serine and threonine residues through activation of protein kinases, including JNK.

How does phosphorylation of BAD by JNK at Thr201 inactivate the proapoptotic molecule BAD? Our results show that phosphorylation of BAD by JNK at Thr201 significantly reduced the association of BAD with BCL-X_L in vitro (Figures 6A and 6B). Furthermore, in cells expressing the constitutively active JNKK2-JNK1, BCL-X_L only associated with nonphosphorylated BAD but not Thr201 phosphorylated BAD (Figure 6C). These data suggest that, like the known serine phosphorylations, Thr201 phosphorylation of BAD also reduces its association with the antiapoptotic molecule BCL-X_L, thereby inhibiting BAD proapoptotic activity.

Where does phosphorylation of BAD by JNK occur? It has been shown that BAD can be phosphorylated on the outer membrane of mitochondria by the mitochondria-anchored PKA (Harada et al., 1999) or Raf-1 (Wang et al., 1996). However, another BAD kinase, Akt, is a cytoplasmic protein kinase (Datta et al., 1997; del Peso et al., 1997). We found that JNK and BAD were in both cytosol and crude membrane preparations (containing mitochondria; Zha et al., 1996). In response to IL-3 withdrawal, a portion of BAD proteins translocated from cytosol to membrane preparations (data not shown; Zha et al., 1996). However, there was no detectable translocation of JNK (data not shown). Thus, it is likely that, like Akt, JNK phosphorylates BAD in the cytosol, thereby preventing BAD translocation.

Unlike the known serine phosphorylation sites (Ser112, Ser136, and Ser155), whose surrounding amino acids comprise a 14-3-3 binding site (typically RSXpS/TXP or RXXXXpS/TXP; Yaffe, 2002), the surrounding amino acids of Thr201 (GKGGSpTPSQ) do not comprise the 14-3-3 binding site. Despite this, the BAD(T201A) mutant was as potent as the BAD(S112A) mutant in promoting IL-3 withdrawal-induced apoptosis (Figure 7F). It is possible that binding to 14-3-3 proteins may not be the sole mechanism for sequestering the proapoptotic molecule BAD in the cytoplasm. Phosphorylation at Thr201 may sequester BAD by targeting it to a yet-to-be identified cytoplasmic protein(s). Since the BAD(S112A/T201A) double mutant was more potent in inducing apoptosis (Figures 7B and 7F), phosphorylation of Thr201 may work in accord with phosphorylation of Ser112 (or other serines) in suppressing the proapoptotic activity of BAD. It is possible that Thr201 phosphorylation of BAD by JNK may participate in the effort of reducing BAD binding affinity to BCL-X_L (Figure 6), perhaps preventing the translocation of BAD to the mitochondria, whereas serine phosphorylations of BAD are responsible for the retention of BAD in the cytosol through binding to 14-3-3. Further studies are needed to test these possibilities.

Concluding Remarks

The role of JNK in apoptosis is highly dependent on the cellular context, such as cell type, the nature of the death stimulus, the duration of its activation, and the activity of other signaling pathways (Lin, 2003). JNK is required for TNF- α -induced apoptosis in the absence of NF- κ B activation or for UV-induced apoptosis. However, the results of this study show that JNK activation is also required for growth factor-mediated cell survival through phosphorylation and inactivation of the proapoptotic Bcl-2 family protein BAD. Future studies on the integration of the cellular signaling network should provide a better understanding of how JNK is involved in a cell's life and death decision making.

Experimental Procedures

Reagents

Antibodies against JNK, p38, ERK, phospho-p38, phospho-ERK, and phospho-BADS112 were from PharMingen or Cell Signaling. Anti-BAD(10929) and anti-BCL-X_L antibodies were gifts from Drs. Stanley Korsmeyer and Larry H. Boise, respectively. The JNK inhibitor SP600125 was synthesized and purified (Tang et al., 2002). Anti-M2 antibody and Hoechst (H33258) were from Sigma. Caspase-3 substrate VII (DEVD-AFC) and caspase-9 substrate (LEHD-CHO) were from Calbiochem. [γ -³²P]ATP and [³⁵S]methionine were from NEN. Anti-phospho-BADT201 antibody was generated by injecting KLH-conjugated phosphopeptide (CRNLGKGGSpTPSQ) into rabbits (Cocalico Biologicals Inc.) and purified.

Plasmids

pGEX-KG-BAD was a gift from Dr. Stanley Korsmeyer. S128A, S136A, and T201A mutations were introduced using the Quick-Change site-directed mutagenesis kit (Stratagene), and mutations were verified by DNA sequencing. pcDNA3.1-Hygro(+) vectors encoding M2-BAD and M2-BAD(T201A) were generated by addition of an N-terminal M2 tag to BAD and confirmed by DNA sequencing. Δ N-BAD(115-204) and Δ C-BAD(1-114) constructs were generated by PCR and subcloned into pGEX-KG between BamHI and HindIII to generate the GST- Δ C-BAD and GST- Δ N-BAD fusion proteins. PSR α -HA-JNKK2-JNK1 and pSR α -HA-JNKK2(KM)-JNK1 have been described previously (Tang et al., 2001a; Zheng et al., 1999).

Cell Culture and Transfection

FL5.12, FL5.12 BCL-X_L/BAD cells were gifts from Dr. Stanley Korsmeyer. FL5.12 BCL-X_L cells were a gift from Dr. Charles Rudin. *Jnk1*^{-/-} MEFs were prepared from *Jnk1* null mice, a gift from Dr. Michael Karin (Sabapathy et al., 1999; generation of *Jnk1* null mice was supported by NIH grant ES04151 to Dr. Michael Karin), and immortalized as described (Tang et al., 2001a). MC/9 and FD-CP-1 cells were gifts from Dr. Vincent Duronio. Cells cultured in ISCOVE'S or RPMI medium, supplemented with 10% Wehi soup, 5% fetal bovine serum (FBS), 2 mM glutamine, 100 U/ml of penicillin and 100 μ g/ml of streptomycin. FL5.12 BCL-X_L/M2-BAD or M2-BAD(T201A) were generated by transfecting FL5.12 BCL-X_L cells with pcDNA3.1-Hygro(+)-M2-BAD wt or T201A (20 μ g each) and selected against Hygromycin B (1 mg/ml).

Protein Kinase Assays and Immunoblotting

Immune complex kinase assays were performed and quantitated as described previously (Lin et al., 1995; Tang et al., 2001a). Active GST-JNK1 was prepared by in vitro activation, in which purified GST-JNK1 proteins were incubated with extracts of cells transfected with HA-MEKK1, purified with GSH column (Lin et al., 1995). HA-Akt was isolated from transfected COS-1 using immunoprecipitation with anti-HA antibody (Santa Cruz). Immunoblotting was performed as described (Tang et al., 2001b).

Phosphoamino Acid Analysis and Two-Dimensional Tryptic Phosphopeptide Mapping

Two-dimensional tryptic phosphopeptide mapping or phosphoamino acid analysis of phosphorylated GST-BAD proteins was done as described (Lin et al., 1992, 1995).

Apoptosis Assays

FL5.12 cells or stable cells were harvested at certain time points. Cells were stained with Hoechst for 1 min, followed by two washes with PBS. After centrifugation, cells were dropped on a cover slide. Nuclear condensation and DNA fragmentation were visualized by fluorescence microscopy. Caspase-3/9 assays were performed as previously described (Tang et al., 2002).

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